

### Apply Scale and Unit Comparison in Real Tasks

### Step 8

Elementary  
(Grades 2–4)



Today I am learning to apply scale and unit comparison to solve real-life problems.

#### 1. Map Scale

The map scale is 1 cm = 10 km.

Use the scale to complete the table.



| Route  | Map distance (cm) | Actual distance (km) |
|--------|-------------------|----------------------|
| A to B | 3 cm              | _____                |
| B to C | 5 cm              | _____                |
| A to C | 8 cm              | _____                |
| C to A | 6 cm              | _____                |

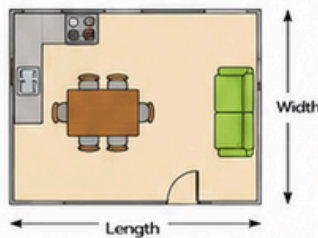
If the actual distance from Town A to Town C is 80 km, what would the map scale be?

1 cm = \_\_\_\_\_ km

#### 3. Scale Drawings

The scale of the drawing is 1 cm = 2 m.

Complete the table.



| Item         | Drawing (cm) | Actual size (m) |
|--------------|--------------|-----------------|
| Length       | 7 cm         | _____           |
| Width        | 4.5 cm       | _____           |
| Table length | 2.5 cm       | _____           |
| Sofa length  | 3 cm         | _____           |
| Door width   | 1 cm         | _____           |

#### 5. Recipes and Scaling

A recipe makes 4 servings.

Use the table to find how much ingredient is needed.

| Ingredient | For 4 servings | For 6 servings | For 8 servings |
|------------|----------------|----------------|----------------|
| Flour      | 2 cups         | _____          | _____          |
| Sugar      | 1 cup          | _____          | _____          |
| Milk       | 1.5 cups       | _____          | _____          |
| Eggs       | 2              | _____          | _____          |

**Remember!**

To scale up or down, multiply by the same amount.



#### 7. Real-Life Problems

- a) A car travels 240 km using 16 litres of fuel. How many litres will it use to travel 360 km?  
Answer: \_\_\_\_\_ litres



- b) A garden is 12 m wide and 18 m long. A path 1.5 m wide runs all around the outside. What is the area of the path?  
Answer: \_\_\_\_\_ m<sup>2</sup>



- c) A packet of 30 stickers costs \$2.70. How much would 45 stickers cost?  
Answer: \$ \_\_\_\_\_



- d) A water tank holds 1500 litres. If it is 62% full, how many litres of water are in the tank?  
Answer: \_\_\_\_\_ litres



#### 2. Unit Conversion

Fill in the missing values.

- a) 5 m = \_\_\_\_\_ cm      f) 4 cm = \_\_\_\_\_ mm  
b) 3 km = \_\_\_\_\_ m      g) 2.6 km = \_\_\_\_\_ m  
c) 2 L = \_\_\_\_\_ mL      h) 3.75 L = \_\_\_\_\_ mL  
d) 750 g = \_\_\_\_\_ kg      i) 1250 g = \_\_\_\_\_ kg  
e) 15 kg = \_\_\_\_\_ g      j) 0.3 m = \_\_\_\_\_ cm

**Tip!**

Remember:

1 m = 100 cm      1 km = 1000 m      1 L = 1000 mL      1 kg = 1000 g

#### 4. Best Buy?

Compare the unit prices to find the best buy.

- a) Milk: Option A: 2 litres for \$3.20      Option B: 3 litres for \$4.50

Best buy: \_\_\_\_\_

- b) Cereal: Option A: 450 g for \$2.70      Option B: 700 g for \$3.85

Best buy: \_\_\_\_\_

#### 6. Painting a Wall

1 litre of paint covers 8 m<sup>2</sup> of wall.

How much paint is needed for each wall?



|    | Wall size (m <sup>2</sup> ) | Calculation | Paint needed (litres) |
|----|-----------------------------|-------------|-----------------------|
| a) | 16 m <sup>2</sup>           | _____       | _____                 |
| b) | 24 m <sup>2</sup>           | _____       | _____                 |
| c) | 36 m <sup>2</sup>           | _____       | _____                 |
| d) | 48 m <sup>2</sup>           | _____       | _____                 |

#### 8. Think and Solve

Use what you know about scale and units to solve.

- a) A model car is 12 cm long. The real car is 4.2 m long. What is the scale of the model?  
Answer: 1 cm = \_\_\_\_\_



- b) A room is 5 m by 4 m on the plan. The plan uses a scale of 1 cm = 1 m. What are the length and width on the plan?  
Length: \_\_\_\_\_ cm      Width: \_\_\_\_\_ cm



- c) A bus can carry 48 passengers. A trip needs 216 passengers. How many buses are needed?  
Answer: \_\_\_\_\_ buses



I can...

- ☐ apply map scales.
- ☐ convert between units.
- ☐ compare unit prices.
- ☐ scale recipes and amounts.
- ☐ solve real problems using scale and units.
- ☐ explain my thinking.



Key Words

- scale** – a way of showing the size of something compared to real life  
**unit** – a standard amount used to measure  
**proportion** – a fair or equal part of a whole  
**convert** – change from one unit to another



Great job!

You are using scale and units to solve real-life problems!